



## Document Management System



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**Title:** Therapeutic Plasma Exchange

Therapeutic plasma exchange (TPE) or plasmapheresis is a procedure which removes circulating antibodies, immune complexes and toxins from the blood. TPE has been successfully used in various pediatric immunological and non immunological diseases.

### Indications:

#### a. Neurology: According to American society for Apheresis (ASFA)

- Guillain-Barre syndrome/ acute inflammatory demyelinating polyneuropathy.
- Acute disseminated encephalomyelitis/ acute transverse myelitis.
- Chronic inflammatory demyelinating polyradiculoneuropathy.
- Myasthenia gravis (in moderate-severe crisis or as preoperative therapy).
- Multiple sclerosis (acute/relapse).
- Pediatric autoimmune neuropsychiatric disorders associated with streptococcal infection.

#### b. Nephrology: According to American society for Apheresis (ASFA)

- ANCA associated rapidly progressive glomerulonephritis.
- Good pasture's syndrome.
- Focal segmental glomerulosclerosis ( if recurrence in transplanted kidney, steroid resistance in native kidney)
- Henoch Schonlein purpura.
- Immunoglobulin A nephropathy ( chronic progressive, crescentic).
- Thrombotic micro-angiopathy, complement mediated.
- Thrombotic micro-angiopathy, drug associated.
- Thrombotic micro-angiopathy, Shiga toxin mediated( severe neurological symptoms, streptococcus pneumonia)

c. Acute Hepatic Failure as adjunctive therapy.

d. Familial hypercholesterolemia.

e. Fulminant Wilson's disease.

Another Classification of indications of TPE

- **To remove**
- **AutoAb (+ probable autoAb):** Thrombotic Thrombocytopenic Purpura (TTP), Immune Thrombocytopenia (ITP), Autoimmune Hemolytic Anemia, Evan's syndrome, Antiphospholipid crisis, Anti-GBM glomerulonephritis (& Goodpasture's), ANCA nephritis, ANCA vasculitis (& Wegener's), MPGN (C3 nephritic factor autoantibody), Acute Guillain-Barré syndrome, Chronic Inflammatory Demyelinating Polyneuropathy (CIDP), Myasthenia Gravis, Lambert-Eaton syndrome, Stiff Person syndrome, Acute Disseminated Encephalitis (ADEM), Anti-NMDA-R Encephalitis, Anti-VGKC (Voltage-gated potassium channel) Limbic Encephalitis, Neuromyelitis Optica (NMO), some Multiple Sclerosis (MS), etc.
- **Antigen-Antibody complexes:** Hepatitis C vasculitis, Lupus vasculitis, etc.
- **AlloAb:** Transplant sensitization, Transplant rejection (humoral), Transfusion reactions, etc.
- **Paraproteins:** Waldenstrom's, Hyperviscosity, Light-chain neuropathy, Light-chain glomerulopathy, Myeloma cast nephropathy, etc.
- **Non-Ig proteins:** Focal Segmental Glomerulosclerosis (FSGS).
- **Endogenous toxins:** Hypercholesterolemia, Liver failure, Sepsis (& SIRS), Phytanic acid (Refsum's disease), etc.
- **Exogenous poisons:** Amanita, drugs, etc.
- **To replenish plasma factors:** TTP, MPGN with inactive Factor H, etc.

### Therapeutic Plasma Exchange:

This procedure involves the removal of large volume of plasma that is replaced by replacement fluid which is either 5% albumin or fresh frozen plasma (FFP).

### Mechanism of action of TPE:

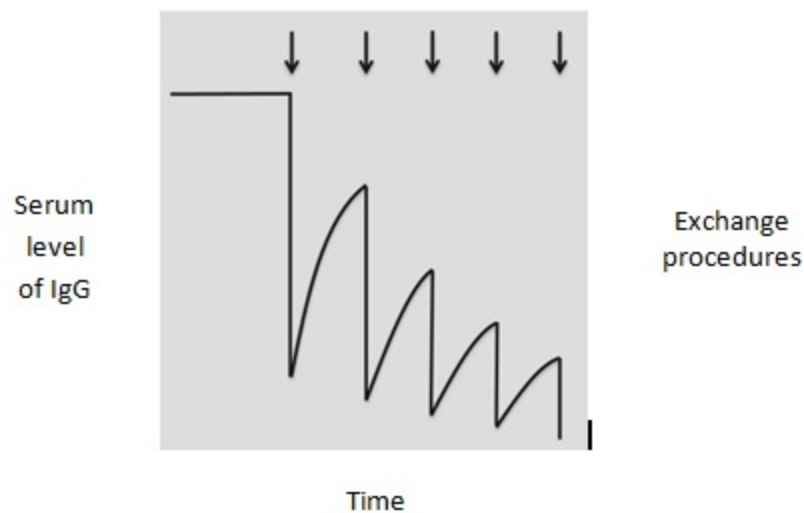
It effectively removes large-molecular weight pathological antibodies, immune complexes, complement components, and coagulation cascade factors.

The clearance of different circulating substances depends not only on their size but also on their distribution between intravascular and extra-vascular compartments. For example IgM is removed more effectively than IgG because it is largely intravascular.

A course of TPE typically comprises 3-5 or more separate exchange procedures undertaken at 24-48hrs intervals. The rationale for this is determined by the removal kinetics of the exchange process.

In each exchange 1-2 total plasma volume are replaced over a period of several hours. Depending on the exponential decay function a single total plasma volume (TPV) exchange will remove 63% of original plasma, 1.5 TPV removes 77.7% and 2 TPV removes 86.5%.

However, for a typical immunoglobulin G- class autoantibody, only around 30% of the total amount in the body is located in the intravascular space, the rest residing in the tissues and interstitial fluid (third space). Multiple exchanges are therefore required; after each exchange a concentration gradient is established between the extra- and intravascular compartments; the interval between exchanges allows time for equilibration of pathogenic molecules across compartments (tissue to serum, and importantly cerebrospinal fluid to serum), often



Idealized kinetics of a plasma exchange course for removal of a typical immunoglobulin (IgG)-class autoantibody.

Additional consideration specific to the pathogenic molecule include its rate of ongoing synthesis and catabolism, volume of distribution and rate of equilibration between bodily compartments which may be as slow as 1-3% per hour for the large molecules. For example IgG and IgA distribute more extensively in the extra-vascular space more than IgM because of their lower molecular mass, hence they re-accumulate in the plasma more readily after each exchange and require a larger number of exchanges to achieve the same clearance. The targeted cumulative effect of a course of exchanges is to remove up to 90% of the total body burden of pathogens.

### Plasma exchange techniques:

In children, continuous centrifugation (in which blood is removed and replaced simultaneously) is generally considered preferable to intermittent centrifugation (in which blood is removed in batches, which are separated and reconstituted individually before replacement).

### Preparation for treatment:

#### A. Venous access (Dialysis line):

The preferable access usually is a dialysis catheter and preferable site is the internal jugular vein especially in small children. Femoral can be considered in older children but the mobility will be limited. Subclavian is not preferred as it has high chance for blockage. For the line size refer to table available at the end of document.

| Wt (kg)  | Dialysis cath size |
|----------|--------------------|
| 2.5-5 kg | 6-7 Fr             |
| 5-8 kg   | 7 Fr               |
| 8-10     | 8 Fr               |
| 10-15    | 9 Fr               |

>30

12 Fr

### B. Blood Investigations:

1. Prior to each exchange the following lab investigations are required; Blood group, CBC, coagulation profile, Ionized calcium, renal function.
2. Serum Albumin: before treatment and at the end of treatment.
3. Immunoglobulin levels: to be done at the beginning of treatment course and repeated after the course to decide of replacement is needed.

### C. Hemodynamic stability and monitoring:

The patient undergoing this treatment should be monitored closely preferable and arterial line should be in place and cardiac monitor also to detect any complications like arrhythmia and hypotension. The machine used for the exchange usually the volume that will be in the extracorporeal set is usually around 160-180ml and this amount is considered to be a significant amount and lead to hemodynamic instability like hypotension and These children might also develop transfusion reactions like fever, rash ..etc. An acceptable extracorporeal blood volume is around 8-10% of total blood volume with absolute maximum of 15% if so then we need to consider either priming with blood/ volume expanders or a side drip of PRBC Tx once starting the treatment to maintain the haemodynamic stability.

### D. Premedications:

If blood products are to be used premedication with Paracetamol and antihistamine may be needed. Steroids might be considered if previous history of transfusion reaction.

### **Therapeutic Plasma Exchange Prescription:**

The standard practice is 1.0-1.5 plasma volume is exchanged per session Can go upto 2 plasma volume also in cases like acute liver failure. For you to calculate the plasma volume we need to calculate the total blood volume. For this there are several formulas and methods for our purpose we will use the following table for blood volume calculation.

| Age group                   | Total blood volume |
|-----------------------------|--------------------|
| Neonate( birth upto 1month) | 85ml/kg            |
| Infant( 1month-1yr)         | 80ml/kg            |
| Child( >1yr-13yr)           | 70ml/kg            |

The plasma volume is calculated as follows: Plasma volume= TBV X(1-Hct/100).

A course of 5 exchange sessions is usually prescribed on alternate day bases.

Setting the plasma exchange machine: You have to set the following:

Blood flow rate is usually 50-60ml/min and this can be adjusted based on the child hemodynamics and stability.

from 1-2 hrs and based on that set the blood flow and the plasma exchange rate.

The machine will also require setting the total plasma volume to be exchanged at the beginning of the treatment.

During each session we should start anticoagulation in the form of Heparin infusion 10IU/kg/hr and monitor the coagulation.

#### Complications of TPE:

1. Hypotension.
2. Hypertension.
3. Symptomatic hypocalcemia.
4. Allergic reactions; urticaria, pruritis.
5. Chills.
6. Gastrointestinal; abdominal pain, vomiting.
7. Bleeding, Hematoma.
8. Line related complication; obstruction, displacement, infection.
9. Depletion of non-pathological blood components; albumin, Immunoglobulins.

For the prescription order sheet refer to the TPE excel sheet available in PICU nursing station computer.

#### Plasma Exchange procedure: A senior physician should be around to run and monitor the exchange session.

1. Fill up the plasma exchange order sheet and calculate the plasma volume to be exchanged. ( to be found in the computer at the nursing station).
2. Decide regarding the use of either FFP or albumin to replace the plasma to exchange. ( based on the coagulation result if normal can use albumin if deranged to replace with FFP.
3. Prime the CRRT machine after setting the plasma exchange kit, we can prime with normal saline and follow the machine instruction until it is ready to be connected to the patient. You will be required to enter the plasma volume to be exchanged into the machine.
4. Access the central line with aseptic technique and precautions.
5. Remove the heparin present within the central line lumens along with extra 2-3ml per lumen.
6. Flush the line with normal saline and make sure about the smooth back flow from the arterial lumen and smooth flushing in the venous lumen.
7. Prepare heparin infusion at a rate of 20IU/kg/hr (higher dose as the blood flow is slower than that required for CRRT) to the patient if the coagulation is normal. If the patient is having deranged coagulation we might consider running the session without heparin.
8. Connect the line to the CRRT machine and follow the instructions in it.
9. The blood flow to be set between a minimum 20-50ml/min and accordingly the plasma exchange rate will be set at 20% of the blood flow rate. We should aim for a 1-2 hrs session. (eg. Plasma volume exchange is 800ml. At a blood flow rate of 50ml/min and plasma exchange rate of 10ml/min it will take around 80min to finish the therapy.)
10. While initiating the therapy we should adjust the blood flow gradually and keep monitoring the hemodynamics of the patient and his stability.
11. We should keep a close eye on the patient for development of any of the known complications of the procedure and act promptly.
12. When the session is ending follow the instructions of the machine regarding termination of the session.
13. Don't forget to use sterile technique while accessing the dialysis line locking the line with heparin accordingly.
14. Do the needful lab works like ABG, CBC, coagulation, UE1 and bone within an hour after terminating the therapy and correct accordingly.

#### Reference

1. Therapeutic plasma exchange in paediatric neurology: a critical review and proposed treatment algorithm: a critical review and proposed treatment algorithm. Micheal Eyre. 2018. Developmental Medicine and child Neurology.
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3. Therapeutic Plasma Exchange UHL Renal Guideline. University Hospitals of Leicester. (NHS). Clinical guideline. 2021.

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